Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**AI-Based Non-Contact Magnetic Sensing Technology for Remote Monitoring of Vessel Engine Start Batteries and EPIRB to Prevent Maritime Distress Accident**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication (√), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical.Electronics		*Information & Communication*	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	*Manufacturing and Measurement/Evaluation Equipment*	*Software*	*Software Solutions*
Concept of Proposed Technology	**Core Concept** A significant number of fishing vessel distress accidents are caused by **battery-related failures rather than mechanical engine faults**. In particular, two critical battery systems play decisive roles in maritime safety: i. **Engine start batteries**, which directly determine whether a vessel can restart its engine during operations ii. **EPIRB batteries**, which determine the reliability of emergency distress signal transmission when an accident occurs The proposed technology introduces an **integrated, non-contact battery monitoring system** that			

Concept of Proposed Technology

simultaneously monitors both battery systems using a unified technological framework. The technology enables **non-contact health monitoring and lifetime estimation of vessel batteries** such as those used in EPIRB or Strating Motor without disassembly or physical electrical connection.

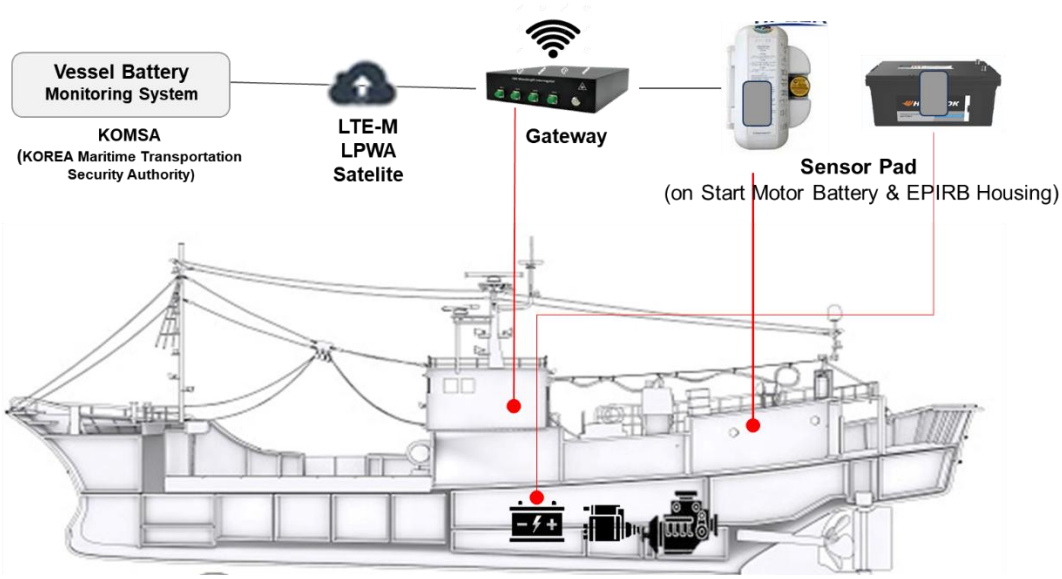
1. Basic Principle

Low-power **Time varying magnetic fields are externally applied** to the vessel battery through the device housing, and the **changes of magnetic flux** are measured using a surface-mounted, ultra-thin electrical coil sensor module. Variations in the resultant magnetic field response characteristics caused by **battery aging, internal chemical degradation, and impedance changes** are analyzed using **AI-based signal interpretation algorithms** to estimate battery State of Charge (SoC) and remaining useful life.

2. System Architecture

The sensor module is attached to the external surface of the start motor battery casing or EPIRB housing and operates **without interfering with the battery connected circuit`s existing functions**. Measured data are transmitted via **wireless communication networks (e.g., LPWA, LTE-M, or satellite communication)** to onboard or shore-based monitoring systems.

By applying artificial intelligence models trained on electromagnetic response patterns under various battery degradation conditions, the system provides **early detection of abnormal battery behavior**, which cannot be reliably identified through conventional voltage-based inspections. This technology allows ship operators to significantly improve maritime safety and maintenance efficiency by performing **remote, continuous, and automated battery inspection used for Engine Start and EPIRB.**



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Rationale for Joint R&D Project	**Critical Technical Barriers** A significant number of medium and small sized vessel distress accidents are not caused by engine mechanical failure, but by unexpected battery discharge. Among these unexpected battery discharge incidents, two battery-related issues described below frequently occur in combination and give fatal influence to marine safety: 1. Failure to restart the engine due to discharge of the engine start battery 2. Unreliable operation of EPIRB caused by battery aging or undetected degradation **1. Failure to restart the engine due to discharge of the engine start battery** On small and medium-sized vessels such as fishing vessels, the engine start battery is typically lead-acid–based systems and are managed independently, often without continuous monitoring or quantitative diagnostics. Since the engine start batteries of the vessels are exposed to harsh marine environments, including vibration, salt corrosion, and irregular charging cycles, lack of engine start battery monitoring often leads to an unexpected engine restart failure during fishing operation. Key limitations of current practices include: i. No continuous monitoring of engine start battery charge status ii. Visual inspection or experience-based judgment by elderly fishermen iii. Lack of early warning before complete battery discharge iv. No digital records of battery condition or usage history When the engine cannot be restarted during fishing operations or anchoring, the vessel becomes disabled and enters a distress situation. Continuous monitoring of battery and early alarming systems to notify battery discharge risk are required to prevent such situations. **2. Unreliable operation of EPIRB caused by battery aging or undetected degradation** EPIRBs on vessels are mandatory safety devices under IMO(International Maritime Organization) and International Convention of SOLAS(Safety Of Life At Sea) regulations and require periodic inspection and battery replacement. However, current inspection methods have several limitations: i. Manual removal and inspection of EPIRBs increase labor, time, and operational costs and especially difficult for aged fisherman who is in charge of EPIRB monitoring and report. ii. Voltage-based inspection methods cannot accurately reflect internal battery degradation iii. Battery condition cannot be monitored continuously during long-term voyages iv. Inspection records are often managed manually, limiting traceability and digital integration **Why Global Collaboration?** This joint R&D project aims to overcome these limitations by integrating: ● **Non-contact magnetic sensing technology** ● **AI-based battery degradation and lifetime prediction algorithms** ● **Maritime-grade wireless communication systems**

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	Purdue University, Collaboration institute of USA, has expertise in detecting battery defects (Li precipitation, metal particles, dendrites, etc.) utilizing magnetic field sensors and magnetic field map analysis. Led by CARES (Center for Advances in Resilient Energy Storage), a research hub jointly established by Purdue University and the Electrochemical Safety Research Institute (ESRI), Purdue conducts systematic fundamental and applied research on the safety, sustainability, and durability of next-generation energy storage systems. CARES covers a wide range of advanced energy storage technologies, including solid-state batteries, metal/liquid electrolyte systems, and sodium-ion batteries, and continuously enhances the industrial applicability of its research outcomes through the operation of application-oriented testbeds. Centered around CARES, Purdue University has established strong industry–academia–government collaboration networks and well-structured technology transfer systems, including the Purdue Research Foundation. Based on this foundation, Purdue has secured numerous core technologies and intellectual property rights in the battery field and maintains strong competitiveness in the commercialization and industrial dissemination of its research outcomes. Collaboration between Purdue university with expertise in non-contact Battery SoC evaluation technology based on magnetic sensing and Korean SMEs with strong capabilities in AI algorithms and system integration based on ICT is essential to accelerate technology maturity and global market entry.
Key Words	**1. Vessel battery** **2. Non-Contact Battery Monitoring** **3. Magnetic Field Sensing** **4. AI-Based Battery Health Diagnosis** **5. Remote monitoring & diagnosis of vessel battery**

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (6)
R&D Objectives	**R&D Objectives** The objectives of this R&D project are as follows: 1. Develop a **non-contact electromagnetic response** model optimized for vessel battery diagnostics 2. Design **AI-based algorithms** for battery degradation assessment and remaining life prediction 3. Develop **a low-power, surface-mounted sensor module** compatible with existing Engine start batteries and EPIRB structures 4. Implement **a wireless remote inspection and monitoring system** for ship-to-shore operation 5. Verify compliance feasibility with **IMO and SOLAS inspection requirements** 6. Verify compliance feasibility with **KOMSA inspection requirements**			

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R&D Target Specifications	• Battery State of Health (SoC) estimation error: within ±10% • Non-contact sensing distance: 5–60 mm • Sensor power consumption: ≤ 50 mW • Wireless data transmission frequency: at least once per day • Environmental reliability: - Salt mist, vibration, and temperature range of –20°C to +60°C
R&D Plan	**Phase 1 (Year 1): Core Technology Validation** • Induced magnetic flux response characteristic analysis • Construction of battery degradation datasets and Development of AI-Model for diagnosis • Design of Remote Battery Monitoring & diagnostic System including Communication Network **Phase 2 (Year 2): System Integration and Prototype Development** • Design of surface-mounted sensor module for Engine Start Battery and EPIRB • Implementation of Remote Engine Start Battery & EPIRB Monitoring & diagnostic System including wireless communication and data acquisition systems • Optimization of AI diagnostic algorithms **Phase 3 (Year 3): Field Demonstration and Verification** • Shipboard and maritime environment testing • Performance comparison with conventional inspection methods • Preparation of certification and commercialization strategies

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Expected R&R of Purdue University** • Development of core magnetic field sensing technologies • Analysis of battery degradation mechanisms • Performance evaluation and technical validation
Expected R&R of Korean SME Partner	**Expected R&R of domestic SMEs.** • Design and Implementation of AI-based diagnostic algorithms • EPIRB structural analysis and sensor integration design • Hardware prototype manufacturing • Design and Construction of remote EPIRB monitoring and diagnosis system including Telecommunication Network • Shipboard testing and field data acquisition • Certification, mass production, and commercialization planning

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Reduction of Fishing Vessel Distress Accidents Caused by Engine Restart Failure** The proposed technology will significantly reduce fishing vessel distress accidents caused by engine restart failures, which often occur due to unstable battery conditions. By enabling real-time monitoring and predictive management of battery performance, operators can prevent unexpected engine shutdowns and restart failures. This will greatly enhance operational stability and reduce emergency situations at sea. **Significant Reduction in EPIRB Inspection Time and Maintenance Cost** The implementation of intelligent battery monitoring systems will streamline the inspection process of Emergency Position Indicating Radio Beacons (EPIRBs). Automated diagnostics and remote condition monitoring will minimize the need for frequent manual inspections and on-site maintenance. As a result, inspection time and maintenance costs can be substantially reduced. **Enhanced Reliability of Maritime Emergency Equipment and Accident Prevention** By continuously assessing battery health and operational status, the reliability of maritime emergency equipment will be significantly improved. Early detection of potential failures enables timely maintenance and replacement before critical malfunctions occur. This proactive approach will play a vital role in preventing maritime accidents and reducing associated risks. **Improved Safety for Fishermen, Particularly Elderly Operators** The proposed system will enhance safety for fishermen by reducing dependence on manual equipment checks and complex maintenance procedures. Elderly operators, who may face difficulties in identifying early signs of battery deterioration, will particularly benefit from automated monitoring and alert functions. This will help create a safer working environment and reduce physical and psychological burdens. **Establishment of Digital, Proactive Battery Management Practices** The project will establish a digital and data-driven battery management framework that enables proactive maintenance and operational decision-making. By analyzing usage patterns and degradation trends, operators can optimize battery replacement cycles and maintenance schedules. This systematic approach will improve equipment longevity and operational efficiency. **Creation of New Smart Maritime Safety Equipment Domestic and Global Markets** The development of intelligent battery management and safety systems will create new opportunities in both domestic and global smart maritime equipment markets. These advanced solutions will meet the growing demand for digitalized and safety-oriented maritime technologies. Consequently, domestic companies will be able to enhance their competitiveness and expand into international markets.

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6. References

(1) IMO SOLAS Convention – EPIRB Inspection and Maintenance Guidelines

(2) Eddy Current Loss Based Non-Intrusive State-of-Charge Estimation System for Lithium Based Batteries-Perdue University